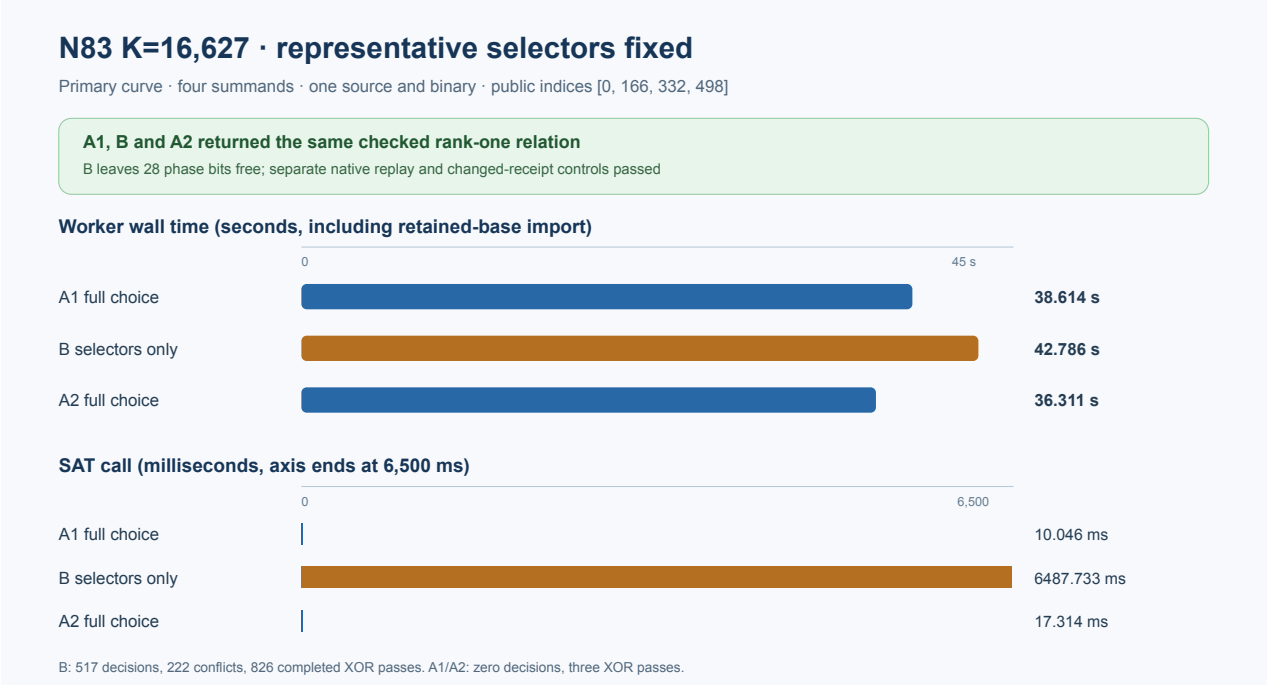
N83 K = 16,627: selector-only four-summand run

Run: 2026-10-18 09:21:26–09:26:49 UTC. **Derived result:** `PASS_THREE_CHECKED_ARMS`. The worker source is committed as `48ebe3b370810e6da2d48e5234e42ba92bba259b`; the retained object, six compiled files, five panel inputs, offline Linux image and binary are bound in `BUILD_RECEIPT.md`. A1/B/A2 used one binary, the same public seven-x-block witness, one CPU, hard 4 GiB, zero swap and a 1,800-second per-arm cap. All exited 0 without OOM.



Common model and measured stages

The pre-pin all-affine chained-S3 model had 201,598 variables, 5,915,238 clauses and 3,901 XOR rows. A1/A2 pinned four selectors and 28 phase-zero wires. B pinned the same four selectors but no phase wires.

Arm	Import ms	Root ms	SAT ms	Wall ms
A1 full	38,117.484	138.582	10.046	38,614.440
B selectors	35,445.011	187.508	6,487.733	42,786.035
A2 full	35,656.408	203.909	17.314	36,311.396

A1 and A2 assigned all 201,598 variables at root, with zero decisions and three completed XOR passes. B assigned 197,586 at root, then returned one SAT model with 517 completed decisions, 222 conflicts, two restarts, 826 XOR passes and 1,124,379 row scans. Its completed XOR-pass timer includes work beyond the separately timed SAT call, so it is not an exclusive CPU fraction. The worker wall includes cold base import; these three L0 observations are not a complete IC timing or robust throughput comparison.

Verification and open work

All three workers checked the original-field S3 chain, exact seven x blocks, stored-point group sum, rank-one row and changed-target rejection. Their row digest is `52b06cef457ecd736ba3f125f4c62bacd31d9bbfc84f2bf6b8d84513f947fe0f`. Separate native replay read the compressed base and each receipt, repeated the field, group and row checks, and rejected one changed-x and one changed-row receipt per arm. The worker did not export SAT assignment bits. The verifier shares the library's field and curve arithmetic and has not run on a second host. B returned before its cap, so conditional phase-only C was omitted.

Free phase wires can be resolved for this constructed x-chain after its representative selectors are supplied. Natural-query relation yield, useful rank, selector acquisition for unknown decompositions, the best prebuilt base at m = 2, 3 or 4, complete cold IC cost and matched rho remain open. `RUN_REPORT.md` links the raw receipts, replay and 70-file hash manifest.